Club Membership Management System

Presented by: Shaik Ushna | N. Pujitha | V. Akshaya Deepika



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Project Overview

Java-Based Console App

Built entirely in Java to manage student clubs through a menu-driven interface

Club & Member Tracking

Handles club creation, student registration, and full membership lifecycle

Capacity Enforcement

Hard limit of 30 members per club with automatic rejection beyond limit

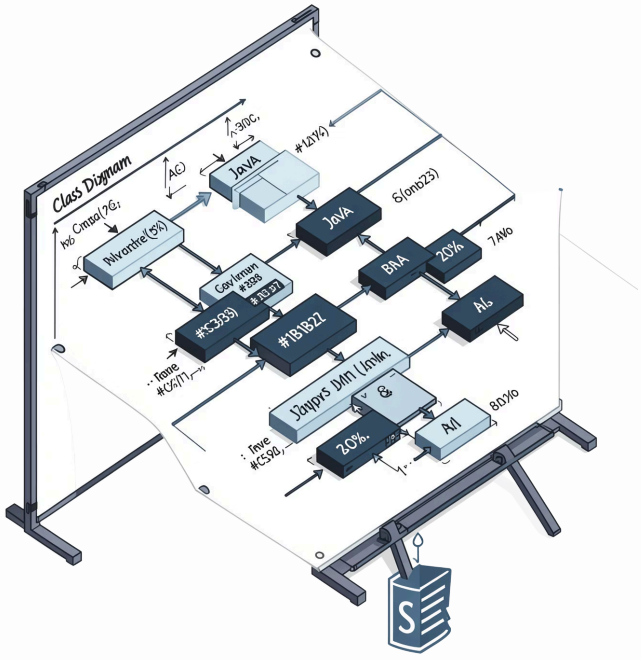
Search & Detection

Partial name search and excess member detection for administrative oversight



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Core Components



Club Class

Stores club ID, name, description, and the list of enrolled members

Student Class

Maintains each student's unique ID and full name for identification

ClubManagement Class

Main controller housing all CRUD operations and system logic

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Club Management Features



Create

Add new clubs with unique IDs and meaningful descriptions



View

Display all clubs with live capacity status: Available, Full, or Excess



Edit

Update club name and description at any time without data loss



Delete

Permanently remove a club and all associated records from the system

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Student & Membership Operations

Student Management

Add Student

Register new students with unique IDs — duplicates are automatically blocked

View Students

List all registered students currently in the system

Membership Operations

Join Club

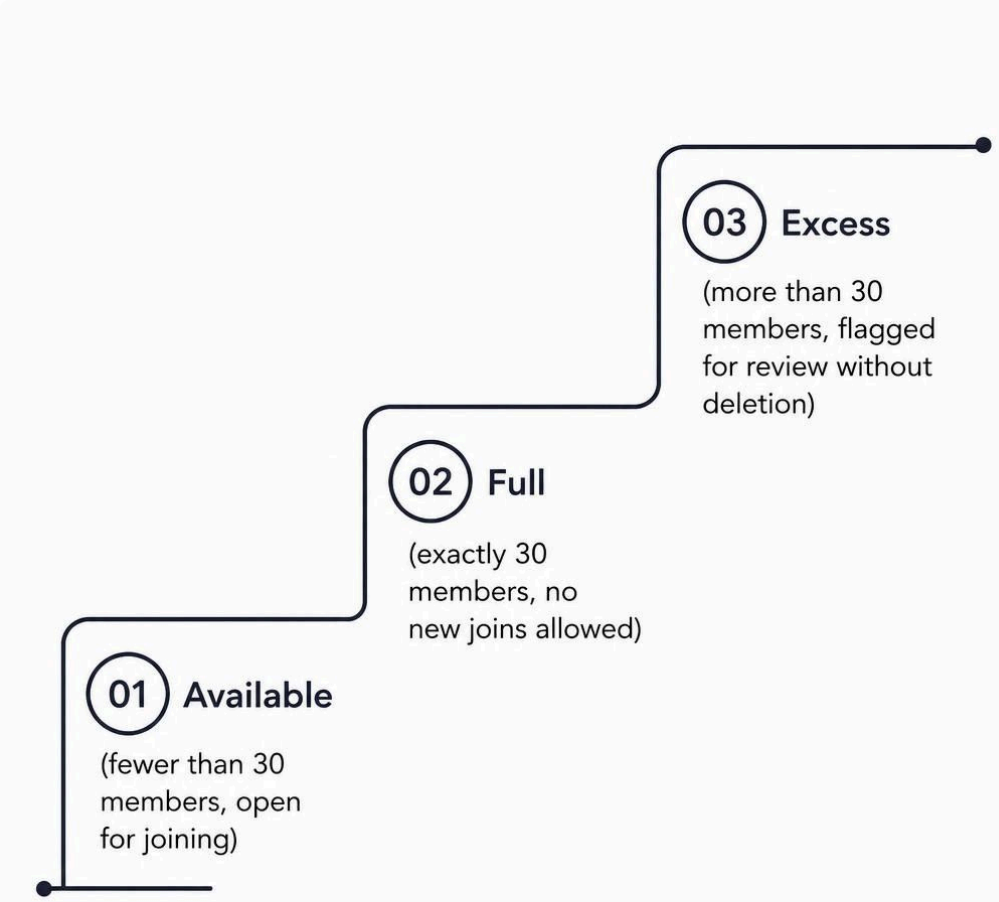
Students request membership; capacity check runs before every join

Display Members

View the full roster of any specific club on demand

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Capacity Management



How It Works

Every club enforces a hard limit of **30 members**. The system evaluates capacity in real time and assigns one of three status labels:

- **Available** — open slots remain, new members may join
- **Full (30/30)** — capacity reached, joins are rejected automatically
- **Excess (>30)** — overcapacity flag raised for admin review

The Excess Check feature flags overcapacity clubs without deleting any data.

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Search & Utility Features

🔍 Search Club

Find any club instantly using a partial name match — fully case-insensitive for ease of use

⚠️ Check Excess

Automatically identifies and lists all clubs whose membership count exceeds the 30-member limit

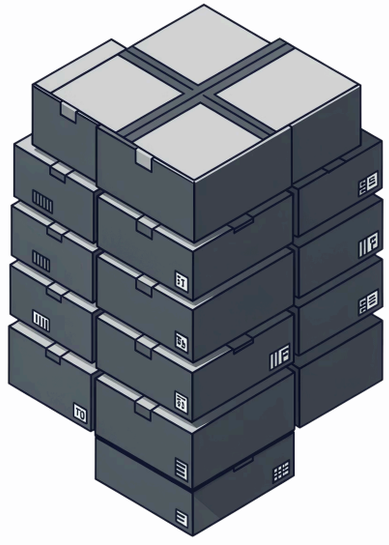
📋 Menu Interface

User-friendly numbered menu guides administrators through every available operation



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Key Data Structures



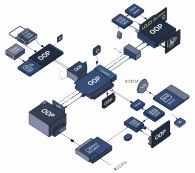
Dynamic Java Collections

The system relies on three core `ArrayList` structures to manage all runtime data efficiently:

1	2	3
ArrayList<Club>	ArrayList<Student>	ArrayList<Integer>
Stores all club objects dynamically — grows and shrinks as clubs are added or removed	Maintains the complete student registry throughout the session	Tracks member IDs within each club instance for fast lookup and validation

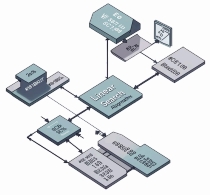
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Technical Highlights



OOP Design

Separate classes for each entity — clean separation of concerns and maintainable architecture



Linear Search

ID lookups use linear search with **O(n)** complexity — simple and effective for the scale



Input Validation

Duplicate ID prevention at both student registration and club creation stages



Capacity Guard

Join-time enforcement ensures the 30-member limit is never silently exceeded

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Thank You

The **Club Membership Management System** demonstrates practical object-oriented programming — combining clean class design, real-time capacity enforcement, and intuitive menu navigation to deliver a fully functional student club platform.

Shaik Ushna

N. Pujitha

V. Akshaya Deepika

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